The Effects of Background Instrumental Music on Human Visual Reaction Time

1. Introduction

Music is commonly played in environments where people study, work, and perform tasks that require concentration. Some people find background music helpful for maintaining focus, while others prefer silence when completing activities that require attention. Although music can influence mood and motivation, its effect on cognitive performance is less straightforward.

One area where this relationship can be examined is visual reaction time. Reaction time measures how quickly an individual responds to a visual stimulus and provides a way to evaluate aspects of attention and information processing. Even small changes in reaction time may be relevant when performing tasks that require individuals to respond quickly and accurately.

The purpose of this experiment was to investigate whether listening to background instrumental music affects human visual reaction time compared with performing the same task in silence. Specifically, the study examined whether participants demonstrated different average reaction times under the two audio conditions.

The research question was:

Does listening to background instrumental music affect human visual reaction time compared with performing the same task in silence?

A two-sided hypothesis test was used because the experiment was designed to determine whether music affected reaction time, without assuming beforehand that it would either improve or worsen performance.

The hypotheses were defined as follows:

- **Null hypothesis (H_0):** There is no difference in mean visual reaction time between the music and silence conditions.
- **Alternative hypothesis (H_1):** There is a difference in mean visual reaction time between the music and silence conditions.

The significance level was set at $\alpha = 0.05$.

2. Experimental Design and Methodology

2.1 Experimental Design

A within-subjects experimental design was used. This means that each participant completed the reaction-time task under both experimental conditions: silence and background instrumental music.

This design was selected because it allows each participant to serve as their own comparison. Reaction time can vary considerably between individuals, so comparing each participant against their own performance helps account for differences in baseline reaction speed.

The experiment included 40 participants. Each participant completed 10 reaction-time trials under each condition, resulting in 20 trials per participant and 800 total observations.

The independent variable was the audio condition, which had two levels:

1. Silence
2. Background instrumental music

The dependent variable was visual reaction time, measured in milliseconds.

2.2 Participant Recruitment

A total of 40 participants were recruited through personal and professional connections. Most participants were coworkers, while the remaining eight were close friends.

Participation was voluntary. All participants completed both experimental conditions using the same testing setup.

Because participants were recruited through convenience sampling rather than random selection from a larger population, the sample may not fully represent the broader population. This is considered when interpreting the results and discussing their generalizability.

2.3 Experimental Procedure

Reaction time was measured using the Human Benchmark Reaction Time Test, available at:

https://humanbenchmark.com/tests/reactiontime

The test measures how quickly a participant responds to a visual stimulus by clicking when the screen changes. Reaction times are recorded in milliseconds.

Each participant completed 10 trials in silence and 10 trials while listening to background instrumental music. The music condition used the same instrumental track, "Africa" by Peter Bence, played through Spotify at a consistent volume for every participant.

To maintain consistency across participants, all trials were completed using the same laptop and mouse. This helped control for potential differences in hardware and input-device performance that could otherwise influence reaction times.

2.4 Experimental Order

To account for potential order effects, participants completed the two conditions in one of two sequences:

- **Order 1:** Silence first, followed by music.
- **Order 2:** Music first, followed by silence.

The participants were evenly divided between the two sequences, with 20 participants in each group.

Using both sequences helped reduce the possibility that completing one condition first would systematically influence performance in the second condition. For example, participants might become more familiar with the reaction-time task after completing their initial trials, or their performance might change due to fatigue.

2.5 Data Collection

Each participant completed 10 reaction-time trials in silence and 10 trials while listening to background instrumental music. Reaction times were recorded in milliseconds and organized into a trial-level dataset.

The dataset contained five variables:

Variable	Description
`participant_id`	Unique identifier assigned to each participant
`condition`	Audio condition: silence or music

`order`	Experimental sequence: silence first or music first
`trial`	Trial number within the condition
`reaction_time_ms`	Recorded reaction time in milliseconds

The resulting dataset contained 800 observations, consisting of 400 trials in each experimental condition.

2.6 Priori Power Analysis

Before conducting the statistical analysis, an a priori power analysis was performed to estimate the number of participants needed for the experiment.

Because the study used a within-subjects design, the power analysis was based on a paired-samples t-test. The calculation used a two-sided significance level of 0.05, a desired statistical power of 80%, and an assumed medium effect size of Cohen's d = 0.50.

The analysis indicated that a minimum of 34 participants would be required to detect an effect of the assumed magnitude with the desired statistical power. The planned sample size was set at 40 participants, providing some additional flexibility beyond the minimum requirement. With 40 participants, the estimated statistical power was approximately 86.9%.

These calculations were performed using Python's Statsmodels library. The assumed effect size was a planning assumption rather than a value derived from the experiment's results.

3. Data Cleaning and Exploratory Analysis

3.1 Data Cleaning

Before performing the statistical analysis, the raw dataset was processed using a Python data-cleaning script.

The cleaning process included checking that all required columns were present, converting variables into their appropriate data types, and identifying missing or invalid values. Additional checks were performed to identify duplicate observations, unexpected condition labels, inconsistent experimental orders, and incorrect participant or trial counts.

The validation process confirmed that the dataset contained 40 participants, with 10 observations per participant in each experimental condition. No missing values or duplicate trial records were identified.

After validation, the cleaned trial-level dataset was saved separately from the raw data. Participant-level averages were also calculated for both conditions.

For each participant, the average reaction time in silence was subtracted from the average reaction time in music:

Paired difference = Music mean reaction time − Silence mean reaction time

A positive paired difference indicates that a participant had a slower average reaction time while listening to music.

These participant-level averages were used for the primary statistical analysis.

3.2 Descriptive Statistics

Descriptive statistics were calculated to summarize reaction-time performance under each condition.

Statistic	Silence	Music
Number of trials	400	400
Mean reaction time	257.71 ms	265.24 ms
Standard deviation	29.58 ms	31.46 ms

The average reaction time was 257.71 milliseconds in silence and 265.24 milliseconds while listening to music.

This represents an average difference of 7.53 milliseconds, with participants responding more slowly in the music condition.

The standard deviations shown above represent variation in the participant-level average reaction times, rather than variation across all individual trials.

The standard deviation of the paired differences was 16.04 milliseconds, indicating the amount of variation in how individual participants responded to the change in audio condition.

3.3 Exploratory Visualizations

Several visualizations were generated to better understand the data before conducting the hypothesis test.

These included reaction-time distributions for both conditions, comparisons of individual participant averages, paired comparisons between conditions, and a visualization of the distribution of paired differences.

The visualizations provided an initial look at the differences between conditions and helped identify patterns that could be relevant to the statistical analysis.

4. Statistical Analysis

4.1 Paired-Samples t-Test

A paired-samples t-test was selected as the primary statistical test because the same 40 participants completed both experimental conditions.

Rather than treating the 800 individual trials as independent observations, the analysis compared the average reaction times for each participant under silence and music. This resulted in 40 paired observations.

The test evaluated whether the population mean paired difference between the two conditions was significantly different from zero.

The paired-samples t-test produced the following results:

Statistic	Result
Number of participants	40
t-statistic	2.9687
Degrees of freedom	39
p-value	0.005093
Mean paired difference	7.53 ms
95% confidence interval	[2.40, 12.66] ms
Cohen's dz	0.4694

The resulting p-value of 0.005093 was below the significance threshold of 0.05. Therefore, the null hypothesis was rejected.

The positive mean difference indicates that participants had slower average reaction times while listening to background instrumental music. The 95% confidence interval suggests that the population mean difference is consistent with music-related increases in reaction time between approximately 2.40 and 12.66 milliseconds, under the assumptions of the paired analysis.

The calculated Cohen's dz of 0.4694 indicates an effect of moderate magnitude using conventional benchmarks.

Overall, the primary statistical analysis provides evidence of a difference in visual reaction time between the silence and music conditions.

4.2 Normality Assessment

One assumption of the paired-samples t-test is that the distribution of paired differences is approximately normal.

A Shapiro-Wilk test was performed on the participant-level paired differences to assess this assumption.

The results were:

- Shapiro-Wilk statistic: 0.9859
- p-value: 0.889792

Because the p-value was greater than 0.05, there was no statistically significant evidence against the normality assumption.

This does not prove that the paired differences follow a normal distribution, but the test did not identify a significant departure from normality. The distribution and Q-Q plot were also examined as part of the diagnostic analysis.

4.3 Wilcoxon Signed-Rank Sensitivity Analysis

As an additional check, a Wilcoxon signed-rank test was performed on the paired observations.

This nonparametric test provides a sensitivity analysis that does not rely on the same normality assumption as the paired-samples t-test.

The test produced:

- Test statistic: 209.0000
- p-value: 0.006096

The p-value was again below 0.05, supporting the conclusion that reaction times differed between the two conditions.

The agreement between the paired-samples t-test and the Wilcoxon signed-rank test provides additional support for the primary finding.

4.4 Exploratory Experimental Order Analysis

An additional exploratory analysis examined whether the experimental sequence was associated with differences in the observed music effect.

For this analysis, the participant-level paired differences were separated into two independent groups based on experimental order:

- Participants who completed silence first.
- Participants who completed music first.

The average paired differences were:

Experimental order	Participants	Mean paired difference	Standard deviation
Silence first	20	2.73 ms	14.32 ms
Music first	20	12.33 ms	16.57 ms

A Welch independent-samples t-test was used to compare the average paired differences between these groups.

The results were:

- Welch t-statistic: -1.9597
- p-value: 0.057546

The p-value was slightly above the 0.05 significance threshold. Therefore, the difference between the two order groups was not statistically significant.

However, the observed difference between the group averages was 9.60 milliseconds. Participants who completed music first demonstrated a larger average difference between conditions than those who completed silence first.

Although this result does not establish a statistically significant order effect, it suggests that experimental sequence may be relevant when interpreting the primary results. Practice effects, fatigue, or other sequence-related factors could potentially contribute to the observed variation.

5. Discussion

The purpose of this experiment was to determine whether background instrumental music affects human visual reaction time compared with silence.

The results provide evidence of a statistically significant difference between the two conditions. Participants had an average reaction time of 257.71 milliseconds in silence and 265.24 milliseconds while listening to music. The resulting paired difference of 7.53 milliseconds indicates that participants responded more slowly, on average, during the music condition.

The paired-samples t-test produced a p-value of 0.005093, providing sufficient evidence to reject the null hypothesis at the 0.05 significance level. The moderate effect size also suggests that the observed difference was not negligible relative to the variation in participant-level differences.

The Wilcoxon signed-rank test produced a similar conclusion, adding support to the primary statistical finding.

One particularly interesting observation was the variation in results across the two experimental sequences. Participants who completed silence first demonstrated an average paired difference of 2.73 milliseconds, while participants who completed music first demonstrated an average difference of 12.33 milliseconds.

Although the difference between these groups did not reach statistical significance, the pattern raises questions about how completing one condition may influence performance in the next. Participants could become more familiar with the task through repeated trials, or their performance could change due to fatigue. Future experiments could investigate these possibilities more directly.

It is also important to distinguish statistical significance from practical significance. Although the difference was statistically significant, the average increase in reaction time was relatively small in absolute terms. The importance of this difference would depend on the specific task and the context in which reaction time is being evaluated.

Overall, the findings suggest that background instrumental music was associated with slower visual reaction times in this particular experiment. However, the results should not be interpreted as evidence that all types of music negatively affect concentration or performance across different activities.

6. Limitations and Future Improvements

Several limitations should be considered when interpreting the findings.

First, the experiment included 40 participants. Although this exceeded the minimum sample size identified by the power analysis, a larger and more diverse participant group could improve the generalizability of the results.

Second, reaction time can be influenced by several factors beyond audio condition, including attention, fatigue, familiarity with the task, and individual differences. While the within-subjects design helped account for differences in baseline reaction speed, these other factors may still have influenced individual performance.

Third, the exploratory order analysis showed a noticeable difference between the two experimental sequences. Although this difference was not statistically significant, it suggests that order-related variation may have affected the results. Future studies could investigate this further by including additional practice trials, standardizing rest periods, or using a larger sample to examine sequence effects more precisely.

Finally, this experiment focused on a specific instrumental music condition and visual reaction-time task. Different music styles, listening volumes, task difficulties, and testing environments could produce different results.

Future research could expand the experiment by comparing multiple music genres, examining different volume levels, or testing whether music affects other forms of cognitive performance, such as memory, attention, or task accuracy.

7. Conclusion

This experiment investigated whether listening to background instrumental music affects human visual reaction time compared with performing the same task in silence.

The results showed that participants had an average reaction time of 257.71 milliseconds in silence and 265.24 milliseconds while listening to music. The paired-samples t-test identified a statistically significant difference of 7.53 milliseconds, with a p-value of 0.005093 and a 95% confidence interval of [2.40, 12.66] milliseconds.

Based on these findings, the null hypothesis was rejected. Participants demonstrated slower average visual reaction times while listening to background instrumental music than while performing the task in silence.

Although the exploratory analysis suggested possible variation related to experimental order, it did not identify a statistically significant order effect at the 0.05 level.

Ultimately, the experiment provides evidence that background instrumental music was associated with a difference in visual reaction time under the conditions studied. It also demonstrates how a controlled experimental design, appropriate statistical testing, and careful interpretation can be used to investigate a question about everyday behavior.